Jason Mac

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Education

University of British Columbia Vancouver, BC, Canada
Bachelor of Applied Science in Computer Engineering *Expected Graduation: April 2028*

- GPA: 88.6 % - Dean's List
- Relevant Coursework: Algorithm Design and Analysis, Database Management Systems, Computer Architecture, Machine Learning, Probability

Douglas College New Westminster, BC, Canada
Engineering Associate *April 2024*

- GPA: 4.24/4.33 - Dean's List
- Relevant Coursework: Object-Oriented C++, Discrete Math

Work Experience

Tenstorrent Toronto, ON, Canada
Software Engineer Intern, AI Compilers *May 2026 – Present*

- Incoming intern on the AI Compiler team, working on C++ graph compilers in MLIR/LLVM-based systems

Projects

Real-Time Chat Application | *Rust, Axum, React, TypeScript, PostgreSQL* GitHub

- Built a full-stack real-time chat application with a Rust backend using Axum and a React TypeScript frontend
- Implemented WebSocket connections for real-time bidirectional messaging using Tokio and a shared users map
- Designed a RESTful API with JWT-based authentication and password hashing via Argon2
- Built a responsive single-page frontend with conversation management, user search, and real-time message delivery
- Deployed the backend on Render and frontend on Vercel with a PostgreSQL database

Lua Bytecode Compiler & Virtual Machine | *C++, Lua* GitHub

- Built a bytecode compiler and stack-based virtual machine for a subset of Lua 5.1 in C++
- Implemented a recursive descent parser producing a AST, converted to bytecode via the visitor pattern
- Designed a bytecode instruction set with support for control flow, local/global variables, and first-class functions
- Implemented lexical scoping, closures, call frames, and native C++ function bindings in the VM runtime

UBC Course Planner | *C#, ASP.NET, .NET MAUI, PostgreSQL* GitHub

- Developed a cross-platform course planning application using .NET MAUI and the MVVM architecture
- Designed a RESTful backend using ASP.NET Core and Entity Framework Core to serve course data
- Built a data client to make HTTP requests to the server and convert JSON responses into domain models
- Implemented ViewModels that asynchronously fetch live course and offering data for the GUI
- Developed an interactive calendar page allowing users to visualize and drag-and-drop course offerings

Tron Light-Cycle Game | *C, RISC-V, Alterra FPGA* GitHub

- Developed a two-player Tron-style light-cycle game in C on a RISC-V softcore running on FPGA hardware
- Rendered graphics directly to a memory-mapped VGA framebuffer, including borders, obstacles, and player trails
- Implemented collision detection by reading VGA pixel values to detect walls, obstacles, and player trails
- Used timer-based interrupts with inline RISC-V assembly to drive game updates
- Handled player input using FPGA KEY-based hardware interrupts, coordinating multiple ISRs via shared state

Technical Skills

Languages: C/C++, C#, Python, TypeScript, Rust, SQL, SystemVerilog
Tools: LLVM, MLIR, Git, Jupyter Notebook, Docker
Libraries/Frameworks: React, .NET, Axum, TailwindCSS, PyTorch